

Before the Breakpoint: detecting pre-resistance from longitudinal MIC distributions in 845,288 bacterial isolates

Mugerwa Moses, Nimusiima Praise

Southern Medical University, China

moses256mugerwa@gmail.com; ninsiimapraise544@gmail.com

Dashboard, Code and Parsed data (GitHub): <https://github.com/Moses256-code/before-the-breakpoint>

Abstract

Antimicrobial resistance (AMR) is a leading cause of death worldwide (Murray et al., 2022), yet routine surveillance reports categorical susceptible, intermediate and resistant breakpoints that signal a problem only after resistance is already clinically established. We propose the fraction of isolates that have crossed the epidemiological cutoff (ECOFF) but not yet the clinical breakpoint, the pre-resistance reservoir, as a leading indicator, and build the Pre-Resistance Alert Score (PRAS) from it. Using 845,288 non-USA Pfizer ATLAS isolates from 82 countries (2004 to 2024), we trained PRAS on 2007 to 2014 and validated it out-of-time on 2015 to 2019. Out-of-time test AUC was 0.90, and leave-one-country-out AUC was 0.92, confirming that the score generalises to countries unseen in training. Among country-pairs whose breakpoint resistance was still below 5%, PRAS separated future crossers from non-crossers at AUC 0.82, against 0.71 for watching the breakpoint rate alone. The signal is mechanism-anchored: the *Klebsiella pneumoniae* ceftazidime-avibactam rise is driven by NDM, which avibactam cannot inhibit. The framework applies to incrementally acquired resistance (7 of 17 pairs tested) and not to single-step resistance. We release an interactive dashboard, the full country-year matrix, and reproducible code.

Objectives

Clinical breakpoints are deliberately conservative and therefore lag the underlying biology (Turnidge & Paterson, 2007), whereas the ECOFF marks the upper edge of the wild-type distribution and flags isolates that have acquired a non-wild-type resistance mechanism (Turnidge, Kahlmeter, & Kronvall, 2006). This work had four objectives:

1. Test whether sub-breakpoint MIC distribution drift predicts subsequent clinical resistance.
2. Build and validate an operational, country-resolved early-warning score across multiple pathogen-drug pairs.
3. Define the scope conditions: which resistance types the framework can detect and which it cannot.
4. Anchor the statistical signal to known molecular resistance mechanisms.

Methods

Data. We used the Pfizer Antimicrobial Testing Leadership and Surveillance (ATLAS) database, accessed through the Vivli AMR Register (Pfizer, 2024), non-USA partition, 2004 to 2024: 845,288 bacterial isolates from 82 countries, with broth-microdilution MIC values for 30 antimicrobials and beta-lactamase genotypes on a subset. Parsing produced 11.4 million isolate-drug MIC observations stored as interval-censored bounds in log₂ space.

MIC handling and quality control. Printed dilution labels carrying censoring qualifiers were mapped to log₂ indices, with panel-edge readings treated as interval-censored. A custom interval-censored estimator recovered known parameters from synthetic data to within 1% at n equal to 5,000. Per-drug panel ranges were checked across years; nine drugs with shifting panels were time-windowed. Colistin was excluded because a 2016 testing standardisation produces artefactual drift, and cefepime was restricted to 2018 onward. Fitting through these methodology breaks would otherwise create spurious drift findings.

Signal and score. For each country-species-drug-year cell with at least 10 isolates, we computed the empirical percentage of isolates above the ECOFF and above the clinical breakpoint, the latter from EUCAST (EUCAST, 2024) and CLSI (CLSI, 2023), with Wilson 95% confidence intervals. PRAS is a logistic regression on five features: current percentage above ECOFF, current percentage above breakpoint, the reservoir (their difference), and the three-year velocity and acceleration of the percentage above ECOFF. The outcome was binary: does the percentage above breakpoint exceed 10% within the next five years. A hierarchical Bayesian version supplied credible intervals.

Validation. Training used predictor years 2007 to 2014 and testing used 2015 to 2019, with non-overlapping outcome windows, giving genuine out-of-time forecasting. We report out-of-time AUC and AUPRC, leave-one-country-out validation (each of 55 countries held out entirely), a feature ablation, calibration, and an analysis restricted to low-baseline cells (percentage above breakpoint below 5%) against a naive baseline that watches the breakpoint rate alone. For *K. pneumoniae* we computed country-year carriage of KPC, NDM, OXA-48, VIM, IMP and CTX-M and aligned it with the PRAS trajectories.

Results

Scope is itself a finding (7 of 17 pairs). Lead-time analysis split the candidate pairs cleanly. The framework applies to carbapenems on Enterobacterales and to novel beta-lactam and beta-lactamase-inhibitor combinations, where resistance accumulates incrementally. It gives no warning for single-step ESBL acquisition, where MICs jump in one move, as in the CTX-M pandemic (Cantón & Coque, 2006), nor for already-saturated pairs. This honest boundary tells surveillance bodies where the tool is worth deploying.

Discrimination and generalisation. Out-of-time test AUC was 0.902 (0.910 Bayesian) with AUPRC 0.85 (Figure 1). Risk buckets were monotonic: cells scoring below 0.10 subsequently crossed 8% of the time, while cells scoring above 0.75 crossed 97% of the time. Leave-one-country-out AUC was 0.920, so the score is not merely memorising country-specific patterns.

The genuine early-warning contribution. The aggregate AUC partly reflects autocorrelation, since a cell already near the threshold trivially crosses it. Restricted to cells where breakpoint resistance was still below 5%, PRAS separated future crossers at AUC 0.82 (leave-one-country-out) against 0.71 for the breakpoint rate alone. A feature ablation localised this advantage to the reservoir level, the percentage above ECOFF, rather than to the trend features. Six representative cells where PRAS was elevated while resistance still looked quiet, and what followed, are shown in Table 1.

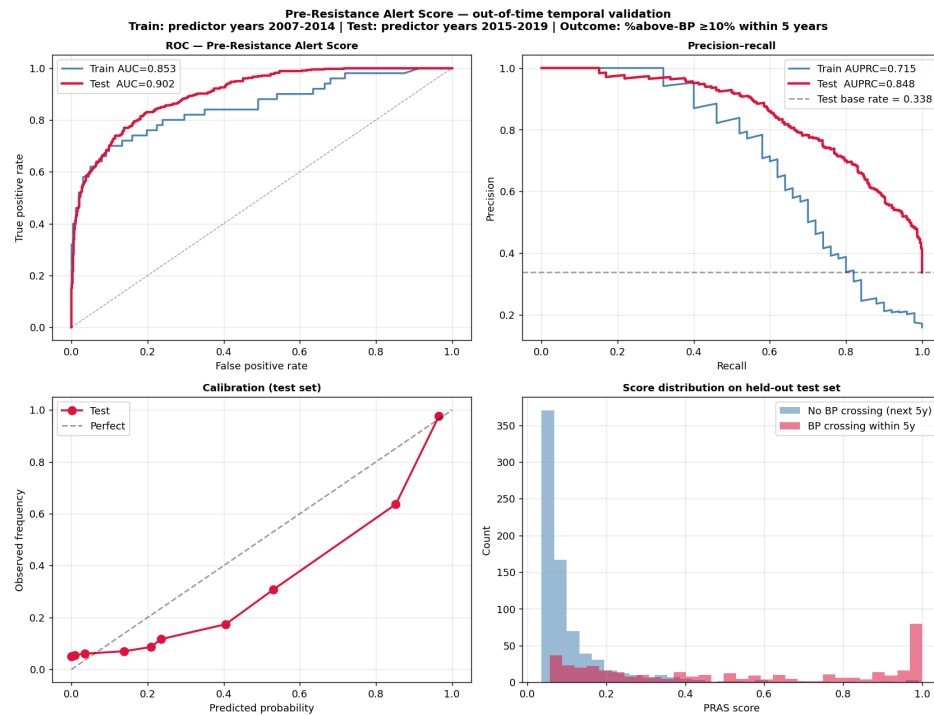


Figure 1. Out-of-time validation of PRAS (train 2007 to 2014, test 2015 to 2019). Test ROC AUC 0.902 and precision-recall AUPRC 0.848; the calibration panel shows the model is conservative in the mid-range; the score-distribution panel shows clean separation between cells that later crossed the breakpoint and those that did not.

Table 1. Worked early-warning examples. In each cell, PRAS was elevated while breakpoint resistance (BP) was still below 5%; resistance then rose sharply within five years.

Country, pathogen-drug, observation year	% above BP then	PRAS	Max % above BP within 5 y
Croatia, <i>K. pneumoniae</i> x meropenem, 2018	4.4%	0.44	41%
Colombia, <i>E. cloacae</i> x meropenem, 2018	3.4%	0.35	73%
Saudi Arabia, <i>K. pneumoniae</i> x meropenem, 2015	3.8%	0.36	33%
Kuwait, <i>K. pneumoniae</i> x meropenem, 2017	4.9%	0.33	39%
China, <i>E. cloacae</i> x meropenem, 2017	3.4%	0.36	18%
Panama, <i>K. pneumoniae</i> x meropenem, 2016	4.5%	0.38	12%

Flagship: K. pneumoniae and ceftazidime-avibactam. A stable distribution broke after 2017, with the percentage above breakpoint rising from below 2% to 12% by 2024. PRAS flagged Greece, Turkey, Brazil and South Africa before their 2018 to 2022 crossings, while Germany, the United Kingdom and France stayed low. The reason is biological: NDM carriage in *K. pneumoniae* rose in parallel, and avibactam does not inhibit NDM, a class B metallo-beta-lactamase (Abboud et al., 2016) (Figure 2).

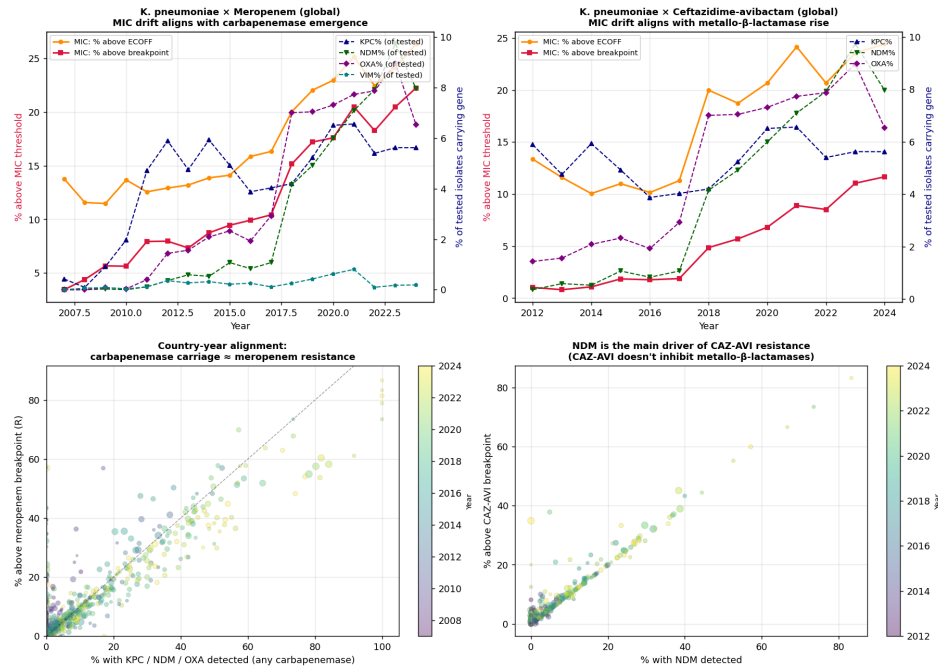


Figure 2. Mechanism anchoring for *K. pneumoniae*. Rising MIC distributions track carbapenemase carriage at the global (top) and country-year (bottom) level. The lower-right panel shows that NDM carriage, not KPC or OXA, drives ceftazidime-avibactam resistance, consistent with avibactam sparing metallo-beta-lactamases.

Calibration and lead-time, stated honestly. The model is well discriminated (Brier score 0.126) but conservative in the mid-range, under-warning rather than over-warning. PRAS does not provide more calendar warning than watching the breakpoint rate approach 5%; its distinct value is discrimination, identifying which still-quiet country-pairs will progress.

Impact of the work

First, surveillance triggers: programmes such as WHO GLASS (World Health Organization, 2022) and ECDC EARS-Net could apply PRAS at the country-pair level to trigger enhanced testing while breakpoint resistance is still low, gaining the discrimination shown here over breakpoint-watching alone. Second, stewardship: hospital and national formularies could be cautious with novel beta-lactam and inhibitor agents in high-PRAS settings even when current resistance is low. Third, methodology preservation: surveillance publishers should keep panel ranges stable across years, because the breaks we identified are lost surveillance signal. Fourth, equity: applied to whatever data exist, the framework places sparse isolates from under-surveilled regions into a global risk

ranking rather than excluding them. Finally, the scope statement is itself a contribution, showing that pre-resistance early warning works for incrementally acquired resistance and that single-step resistance requires complementary genotype-based surveillance.

References

- Abboud, M. I., Damblon, C., Brem, J., Smargiasso, N., Mercuri, P., Gilbert, B., Rydzik, A. M., Claridge, T. D. W., Schofield, C. J., & Frère, J.-M. (2016). Interaction of avibactam with class B metallo- β -lactamases. *Antimicrobial Agents and Chemotherapy*, 60(10), 5655–5662. <https://doi.org/10.1128/AAC.00897-16>
- Cantón, R., & Coque, T. M. (2006). The CTX-M β -lactamase pandemic. *Current Opinion in Microbiology*, 9(5), 466–475.
- Clinical and Laboratory Standards Institute. (2023). *Performance standards for antimicrobial susceptibility testing* (33rd ed., CLSI supplement M100). CLSI.
- European Committee on Antimicrobial Susceptibility Testing. (2024). *Breakpoint tables for interpretation of MICs and zone diameters* (Version 14.0). EUCAST.
- Murray, C. J. L., Ikuta, K. S., Sharara, F., Swetschinski, L., Robles Aguilar, G., Gray, A., ... Naghavi, M. (2022). Global burden of bacterial antimicrobial resistance in 2019: A systematic analysis. *The Lancet*, 399(10325), 629–655. [https://doi.org/10.1016/S0140-6736\(21\)02724-0](https://doi.org/10.1016/S0140-6736(21)02724-0)
- Pfizer. (2024). *Antimicrobial Testing Leadership and Surveillance (ATLAS) database* [Data set]. Vivli AMR Register. <https://amr.vivli.org>
- Turnidge, J., Kahlmeter, G., & Kronvall, G. (2006). Statistical characterisation of bacterial wild-type MIC value distributions and the determination of epidemiological cut-off values. *Clinical Microbiology and Infection*, 12(5), 418–425. <https://doi.org/10.1111/j.1469-0691.2006.01377.x>
- Turnidge, J., & Paterson, D. L. (2007). Setting and revising antibacterial susceptibility breakpoints. *Clinical Microbiology Reviews*, 20(3), 391–408. <https://doi.org/10.1128/CMR.00047-06>
- World Health Organization. (2022). *Global Antimicrobial Resistance and Use Surveillance System (GLASS) report 2022*. World Health Organization.